

BALU NIJAMPATNAM

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Education

Indian Institute of Information Technology Jabalpur

Bachelor of Technology in Computer Science and Engineering — CPI: 9.0

Jabalpur, Madhya Pradesh

Aug. 2024 – Present

Sri Chaitanya Junior College

Intermediate (Class XII) — 982/1000 (98.2 %)

Vijayawada, Andhra Pradesh

2022 – 2024

Sri Chaitanya School

Class X (SSC) — 577/600 (96.1 %)

Tenali, Andhra Pradesh

2021 – 2022

Technical Skills

Languages: C++, C, Python, JavaScript

Web Technologies: React.js, Node.js, HTML, CSS

Databases: MySQL, MongoDB

Libraries & Frameworks: OpenCV, NumPy, Pandas

Developer Tools: VS Code, Vercel, GitHub, Git

Relevant Coursework: Operating Systems, Computer Networks, Probability, Linear Algebra, Database Management Systems, Algorithm Design, Data Structures

Projects

Crowd Management System | [GitHub](#)

April 2026

Python, YOLOv8, OpenCV, Node.js, MongoDB, React

- Developed a real-time crowd management platform that processes CCTV/video streams to monitor crowd density, identify congestion-prone areas, and provide live analytics through an interactive dashboard.
- Integrated the YOLOv8n (Nano) deep learning object detection model with OpenCV to perform person detection and counting on live CCTV feeds, enabling real-time crowd estimation and occupancy monitoring.
- Built a full-stack analytics pipeline using Python, Node.js, MongoDB, and React to store historical crowd data, generate congestion alerts, and visualize crowd trends through REST APIs.

TradeX – Online Trading Exchange Simulator | [GitHub](#)

May 2026

C++, Node.js, Express.js, JavaScript, HTML, CSS

- Engineered a stock exchange simulator that enables user authentication, portfolio management, order placement, and persistent trade tracking, replicating core exchange workflows.
- Implemented a price-time priority matching engine in C++ capable of automatic trade execution, partial order fills, wallet settlement, and real-time holdings updates similar to modern exchanges.
- Developed a responsive trading dashboard featuring live order books, portfolio analytics, trade history, and REST API-based communication between frontend and backend services.

Smart Attendance Tracker | [GitHub](#)

February 2026

HTML, CSS, JavaScript, Local Storage

- Designed and implemented an attendance tracking system that enables students to manage semester-wise subjects, monitor attendance records, and maintain historical academic data.
- Implemented attendance analytics features including percentage calculation, attendance forecasting, required classes to achieve target attendance, and allowable class-skip prediction.
- Created an interactive dashboard with calendar-based attendance history, weekly insights, and local storage-based backup and recovery for seamless data persistence across sessions.

Coding Profiles

LeetCode: [balu_20](#) | 350+ Problems Solved | Rating: 1666 (Max: 1720)

CodeChef: [balu_20](#) | Rating: 1471 (Max: 1471)

Codeforces: [Balu_09](#) | Rating: 1132 (Max: 1132)

Achievements

- Secured Global Rank 3763 out of 42,306 participants (Top 8.8 %) in LeetCode Weekly Contest 492.
- Achieved Rank 347 in CodeChef Starters 237, competing against thousands of programmers.
- Showed sustained academic improvement by increasing SPI from 9.2 in Semester 3 to 9.4 in Semester 4, while maintaining an overall CPI of 9.0.